eman ta zabal zazu

Universidad del País Vasco
Euskal Herriko Unibertsitatea

Retrieval-augmented clinical chatbot for Spanish and Basque

Author: Iker Gutierrez Fandiño

First supervisor: Dr. Ander Barrena Madinabeitia

Second supervisor: Dr. Eneko Agirre Bengoa

hap/lap

Hizkuntzaren Azterketa eta Prozesamendua
Language Analysis and Processing

Master Thesis

September 2026

Departments: Computer Systems and Languages, Computational Architectures and Technologies, Computational Science and Artificial Intelligence, Basque Language and Communication, Communications Engineer.

Laburpena
Abstract in Basque.

Abstract
Abstract in English.

Contents

List of Figures

List of Tables

1 Project definition

1.1 Motivation

Clinical question answering in low-resource languages such as Basque (Euskera) remains largely unsolved. Even for Spanish, the availability of biomedical language models and domain-specific benchmarks is limited compared to English. This project builds and evaluates a retrieval-augmented generation (RAG) pipeline for clinical chatbot applications in both Spanish and Basque, targeting two complementary tasks: open-answer clinical questions drawn from the SNS1064 corpus, and multiple-choice medical exam questions from the CasiMédicos-Arg dataset (Sviridova et al., 2024).

1.2 Objectives

The main goal of this thesis is to determine which combination of retrieval strategy, language model, and inference-time reasoning technique yields the best performance on Spanish and Basque clinical question answering. The specific objectives are:

1. Design and implement an extractive-format RAG pipeline compatible with multiple generator models.
2. Compare dense retrieval (multilingual E5) with cross-encoder reranking (multilingual MS MARCO) across different top-k settings.
3. Evaluate the effect of few-shot prompting (3-shot) combined with retrieval.
4. Assess the benefit of self-feedback (a second LLM call that revises the initial answer) across all conditions.
5. For Qwen3.5-9B, evaluate the additional effect of chain-of-thought thinking mode.
6. Compare a general-purpose multilingual model (Mistral-7B-Instruct, Qwen3.5-9B) with a Basque-adapted model (Latxa-Llama-3.1-8B-Instruct) for Basque tasks.

1.3 Hypothesis

We hypothesise that combining dense retrieval with cross-encoder reranking and few-shot prompting will consistently outperform retrieval-free baselines, and that Basque-adapted models will outperform generic multilingual models on Basque clinical tasks despite a smaller parameter count.

2 System description

2.1 Retrieval pipeline

Documents are indexed using FAISS with multilingual-E5-large embeddings. At inference time the top-k candidates are optionally re-ranked by a multilingual MS MARCO cross-encoder. Retrieved passages are concatenated into the prompt context.

2.2 Generation

Three generator models are evaluated: **Mistral-7B-Instruct-v0.3** (Spanish experiments), **Qwen3.5-9B** (Spanish, with optional thinking mode), and **Latxa-Llama-3.1-8B-Instruct** paired with **Llama-3.1-8B-Instruct** (Basque experiments). All models use an extractive-format prompt that instructs the model to answer using only information present in the retrieved context.

2.3 Self-feedback

Self-feedback (SF) is a two-call procedure: the model first generates an answer, then receives a second prompt asking it to critique and revise that answer. The revised answer is used for evaluation.

2.4 Evaluation metrics

Answers are evaluated against gold references using: ROUGE-L F1 (lexical overlap), cosine similarity of multilingual-E5-large embeddings (semantic overlap), and BERTScore F1 with `bert-base-multilingual-cased` (contextualised token matching). All scores are reported as percentages out of 100.

3 Related work

3.1 Retrieval-augmented generation

Retrieval-augmented generation (RAG) augments language models with external knowledge retrieved at inference time, following a "retrieve-then-read" paradigm (Liang et al., 2025). Early systems relied on sparse retrievers such as BM25, which were later superseded by dense retrievers like DPR and domain-specific models such as MedCPT (Liang et al., 2025). Advanced RAG systems extend this with multi-round structures including iterative, recursive, and adaptive retrieval. More recently, agentic RAG frameworks treat retrieval and chain-of-thought reasoning as actions invocable by the model based on its own metacognitive assessment of when external knowledge is needed (Liang et al., 2025).

A key finding across multiple studies is that RAG does not unconditionally improve performance: irrelevant or noisy retrieved passages can distract models and sustain hallucinations, an effect that is especially pronounced in smaller models (Kim et al., 2025). Kim et al. (2025) conducted the first large-scale expert evaluation of medical RAG, with over 80,000 annotations across 200 queries, and found that simple mitigation strategies such as evidence filtering and query reformulation substantially recover the lost performance. Complementing this, Ozaki et al. (2025) show that models can exhibit calibrated confidence about whether retrieved documents are relevant, suggesting that models possess some capacity for self-assessment of retrieval quality.

3.2 RAG for medicine

Medical RAG has been benchmarked systematically by Xiong et al. (2024), who introduced MIRAGE — a benchmark of 7,663 questions from five medical QA datasets — and showed that MedRAG improves LLM accuracy by up to 18% over chain-of-thought prompting alone. Their study identified a log-linear scaling effect and a "lost-in-the-middle" phenomenon in medical RAG, whereby evidence placed in the middle of a long context is less effectively utilised. Sivakumar et al. (2026) extend this line of work with RAG-X, a diagnostic framework that evaluates the retriever and generator independently across information extraction, short-answer generation, and multiple-choice QA, introducing Context Utilisation Efficiency (CUE) metrics to isolate verified grounding from coincidental accuracy.

Patient-specific knowledge poses an additional challenge: Liang et al. (2025) propose RGAR, which retrieves both factual knowledge from electronic health records and conceptual knowledge from a general corpus simultaneously, consistently improving generation quality and clinical relevance compared to single-source retrieval.

3.3 Self-feedback and self-reflection

Self-RAG (Asai et al., 2024) introduced the idea of training a model to adaptively retrieve passages on demand and critique its own generation using special reflection tokens, achieving strong performance across a diverse set of tasks without requiring task-specific fine-tuning. The self-feedback mechanism in our system is a simpler, inference-only variant: the model generates an initial answer and then receives a second prompt asking it to revise it, without any training signal. A related application appears in OpenScholar (Asai et al., 2026), where a self-feedback-guided generation mechanism enables iterative refinement of long-form scientific synthesis, outperforming GPT-4o by 6.1% on a multi-paper synthesis benchmark while producing far fewer hallucinated citations.

4 The SNS-1064 Dataset

Question answering (QA) systems for the clinical domain require structured and high-quality data to support reliable retrieval and generation. This section presents the SNS-1064 dataset, designed for RAG-based clinical QA tasks. The dataset is derived from Spanish-language clinical guidebooks written by medical experts. Clinical guidebooks consist of "a set of recommendations based on a systematic review of the evidence and an assessment of the risks and benefits of different options, with the aim of optimizing patient care" (GuíaSalud, 2026).

The main objective of this dataset is to transform unstructured clinical text into structured QA instances that separate questions, answers, evidence, and interpretative considerations. This structure is intended to improve downstream retrieval and reasoning capabilities in clinical NLP systems. The dataset and code are publicly available on GitHub: `https://github.com/iker-gutierrez/SNS-1064-dataset`.

4.1 Dataset characteristics

4.1.1 Qualitative description

The dataset is constructed from 6 clinical guidebooks published by the SNS (*Sistema Nacional de Salud* / Spanish National Health System). Each guidebook covers a different clinical domain:

Guidebook	**Source**
Anxiety	Ministerio de Sanidad (2024a)
Diabetes	Ministerio de Sanidad (2026)
Ictus management	Ministerio de Sanidad (2024b)
Palliative care	Ministerio de Sanidad (2021)
Pediatric palliative care	Ministerio de Sanidad (2022)
Secondary ictus prevention	Ministerio de Sanidad (2023)

Table 1: Clinical guidebooks included in the SNS-1064 dataset and their corresponding sources.

The dataset schema is presented in Table 2. Each instance consists of a clinical question, an optional subquestion, a short judgement, supporting evidence extracted from the guideline, and optional interpretative considerations. This structure enables a clear separation between answers (*judgement* column) and their justification (*evidence* and *considerations* columns):

#	**Column**	**Description**
0	guidebook	File name of the guidebook
1	topic	Clinical topic providing context
2	question	Main clinical question
3	subquestion	Refinement of the question (optional)
4	judgement	Short answer to the *(sub)question*
5	evidence	Supporting scientific evidence
6	considerations	Interpretative notes (optional)

Table 2: Schema of the SNS-1064 dataset.

For RAG-based systems, the columns of the dataset can be employed as follows:

- **Useful metadata**: guidebook.
- **Model input**: topic, question, subquestion.
- **Model output**: judgement, evidence, considerations.

4.1.2 Quantitative description

Table 3 summarises the size of the dataset in terms of guidebooks, samples, sentences, and word tokens:

Data type	**Quantity**
Clinical guidebooks	6
Samples	1,064
Sentences	8,602
Word tokens	156,202

Table 3: Total quantities of the SNS-1064 dataset.

Table 4 shows the distribution of samples across the six source guidebooks, illustrating the coverage of different clinical domains.

Guidebook	**Samples**
Diabetes	329
Ictus management	254
Anxiety	248
Palliative care	111
Secondary ictus prevention	98
Pediatric palliative care	24

Table 4: Number of samples per guidebook.

In addition to global counts, per-feature statistics are reported in Table 5. *Questions* and *subquestions* are relatively short (means of 10.2 and 8.5 tokens, respectively), while *judgements* are even more concise on average (5.7 tokens), although they exhibit high variance. In contrast, *evidence* and *considerations* are notably the longest, with mean lengths of 75.6 and 108.0 tokens and large standard deviations, indicating considerable variability. This pattern reflects the structure of clinical guidelines, where concise questions and short answers are accompanied by detailed evidence-based explanations and extended interpretative notes.

Feature	**Mean length**	**Std. dev.**
question	10.2	2.0
subquestion	8.4	3.2
judgement	3.9	7.0
evidence	**73.2**	126.3
considerations	**85.8**	187.6

Table 5: Length statistics (in tokens) for each feature of the SNS-1064 dataset.

4.2 Dataset construction pipeline

The dataset is generated using a rule-based Python pipeline that processes raw text and extracts structured text using regular expressions. This approach has the advantage of being efficient (no GPU needed) and fully deterministic (no Machine Learning, no possible hallucinations). Hallucinations can be especially harmful in the clinical domain.

An overview of the dataset construction pipeline is illustrated in Figure 1:

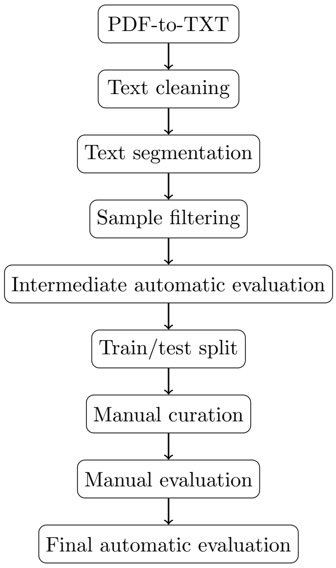

Figure 1: Overview of the SNS-1064 dataset construction pipeline.

4.2.1 PDF-to-TXT

PDF24 (Geek Software GmbH, 2026) was employed to convert each of the clinical guidebooks from PDF format into TXT format. Although this introduces minor formatting noise (e.g. line breaks, headers, and pagination artifacts), these issues are addressed in the cleaning stage.

4.2.2 Text cleaning

Each line extracted from the TXT files is normalised using a dedicated cleaning function that performs the following operations:

- Initial whitespace stripping.
- Bullet-like character removal: {•, ●, −, ∗}.
- Final whitespace stripping.

This step ensures that the text segmentation operates on uniform textual units, reducing noise introduced by PDF extraction artifacts.

4.2.3 Text segmentation

The core of the pipeline is a rule-based text segmentation built using regular expressions. The text is processed sequentially, assigning each line to a structural component via pattern matching. The segmentation logic is composed of the following patterns to identify input features (Table 6) and output features (Table 7):

Input feature	Patterns	e.g.
topic	After "Pregunta" After "Pregunta para responder" After "Subpregunta" Before "Contexto"	
question	`^[a-z])`	`a)`
subquestion	`^[a-z]\.\d+\.`	`a.1.`

Table 6: Text segmentation patterns for input features.

Output feature	Patterns
judgement	After "Juicio:"
evidence	After "Evidencia procedente de la investigación:"
considerations	After "Consideraciones adicionales:" After "Información adicional:"

Table 7: Text segmentation patterns for output features.

Input features are derived from header and pattern-based markers, which define sample boundaries and control when instances are flushed. Output features are activated by keyword triggers and accumulate text until the trigger of the next feature. This mechanism enables reconstruction of a hierarchical structure (*topic* → *question* → *subquestion* → *answers*) from flat text.

4.2.4 Sample filtering

After extraction, a filtering step was applied to remove noisy samples. In particular, 70 instances in the dataset contain only placeholder references in the *evidence* feature: *Ver apartado anterior* ("See previous section") or *Ver apartados anteriores* ("See previous sections"). These cases were removed because they do not provide any bibliographical source but instead refer to previous *evidence* cells. Keeping these placeholder instances would imply associating a single bibliographical source to several clinical questions, which may blur the semantic distance among different QA instances. A short experiment confirms that RAG performance improves by 2.4% when such instances are removed:

Data type	**Unfiltered**	**Filtered**	**Δ**
judgement	84.5	85.3	+0.8
evidence	81.0	81.8	+0.8
considerations	79.7	85.2	**+5.5**
overall	81.7	84.1	**+2.4**

Table 8: Cosine similarity (F1) results with and without sample filtering.

4.2.5 Intermediate automatic evaluation

An intermediate quality check was performed to measure structural consistency. Table 9 shows the presence rates of optional features. *Subquestions* are infrequent, whereas *considerations* are present in the majority of samples.

Feature	**Non-empty cells (%)**
subquestion	4.51
considerations	64.85

Table 9: Presence rates of optional features in the intermediate dataset.

Table 10 shows that the extraction pipeline achieves near-complete presence rates for mandatory features, with noise affecting the *judgement* and *evidence* features.

Feature	**Non-empty cells (%)**
guidebook	100.00
topic	100.00
question	100.00
judgement	**99.91**
evidence	**98.97**

Table 10: Presence rates of mandatory features in the intermediate dataset.

Eleven mandatory cells from *judgement* and *evidence* columns were identified as empty. These samples were flagged for inspection and correction during the manual curation stage.

4.2.6 Train/test split

The dataset is divided into train and test sets with an 80%–20% split, stratified by *guidebook*. Stratification preserves the original proportions of samples per guidebook and accordingly ensures there will be samples from all guidebooks in both sets. Table 11 shows the resulting data distribution:

Set	**Samples**	**Percentage**
Train	851	80%
Test	213	20%

Table 11: Train/test split of the SNS-1064 dataset.

4.2.7 Manual curation

For a reliable evaluation of clinical RAG models, the test set should be noiseless. Hence, a manual curation process was performed on the entire test set (`test_df`). The training set (`train_df`) was only partially inspected, as noise is less critical in training data and full manual curation would be highly time-consuming. The goal of the manual curation step was to fix segmentation errors and guarantee that each sample correctly reflects the intended structure of the source documents.

4.2.8 Manual evaluation

The manual curation concomitantly allowed for a qualitative evaluation of the rule-based pipeline. Overall, the regular expressions for text cleaning and segmentation performed effectively. The majority of samples were correctly structured, but a couple of systematic errors were identified:

- Page headers and footers occasionally remained in the extracted text due to limitations of the PDF-to-TXT conversion.
- In some cases, the *considerations* information was included in the *evidence* feature, evincing that the regex for *considerations* did not catch all text segmentations properly. However, this is a minor issue, since both features explain the *judgement*.

4.2.9 Final automatic evaluation

After manual curation, a final automatic evaluation was conducted to verify that all corrections had been successfully incorporated. Results show a fully consistent dataset with respect to mandatory features (Table 12).

Feature	**Non-empty cells (%)**
guidebook	100.00
topic	100.00
question	100.00
judgement	**100.00**
evidence	**100.00**

Table 12: Presence rates of mandatory features after manual curation.

Table 13 compares optional feature distributions before and after manual curation. *Subquestion* remains virtually unchanged (+0.10), while *considerations* exhibits a more notable increase (+1.50), indicating that curation primarily addressed structural issues without substantially modifying annotation frequencies.

Feature	**Non-empty cells (%)**	Δ
subquestion	4.61	+0.10
considerations	66.35	**+1.50**

Table 13: Presence rates of optional features after manual curation.

4.3 Applicability to RAG-based clinical QA

The SNS-1064 dataset is designed to support RAG systems in the clinical domain. By explicitly separating answers (`judgement` column) from scientific evidence (`evidence` column) and additional considerations (`considerations` column), the dataset enables:

- Improved grounding of generated answers in clinical text.
- Better retrieval alignment between questions and evidence passages.
- Enhanced interpretability through structured reasoning components.

This structure is particularly relevant for high-stakes domains such as healthcare, where factuality, explainability, and traceability of model outputs are essential.

5 Evaluation datasets

Three datasets are used for evaluation, each targeting a different QA format. All three are available in both Spanish (original) and Basque (machine-translated):

- **SNS-1064** (described in section 4): open-answer clinical QA derived from Spanish clinical guidebooks. Questions require free-form answers grounded in evidence passages, making it the primary benchmark for assessing retrieval quality and generation faithfulness.
- **CasiMédicos-Arg** (Sviridova et al., 2024): a multilingual medical QA dataset where clinicians' diagnoses are paired with natural language explanations annotated with argumentative structures. Questions are multiple-choice clinical cases; answers are short option labels, so semantic and contextualised metrics are more informative than lexical overlap for this dataset.

- **Mixed**: a combination of SNS-1064 and CasiMédicos-Arg instances, providing a single aggregate view across both QA formats and enabling a unified comparison of system configurations.

The Basque versions of all three datasets were obtained by translating the Spanish originals, allowing direct cross-lingual comparison under otherwise identical conditions.

6 Dev ablation results

Each table below reports three quality metrics on the overall section (answer + evidence combined): ROUGE-L F1 (lexical), cosine similarity (Cosim, semantic), and BERTScore F1 (BERT-F1, contextualised). Cost is reported as seconds per sample (sec) and total tokens per sample (tok). Yellow rows mark the best configuration per model; ⇒ marks the single best overall. Full tables with all eight quality metrics are available separately. Row labels follow the pattern Xa/Xb: a = Mistral-7B-Instruct (noSF/SF/Δ), b = Qwen3.5-9B (no_think×noSF, no_think×SF, think×noSF, think×SF).

6.1 Spanish results

Table 14 covers the SNS1064 dev set (open-answer, 106 examples). This is the primary Spanish benchmark: answers are free-form clinical text, making lexical and semantic overlap metrics jointly informative.

Table 14: Spanish SNS1064 dev ablation (106 examples, open-answer). Quality on overall section: ROUGE-L, cosine similarity (Cosim), BERTScore F1 (BERT-F1). Cost: seconds per sample (sec), total tokens per sample (tok). Yellow = best per model; ⇒ = best overall.

#	Model	Experiment	Think	SF	Quality ↑			Cost ↓	
					ROUGE-L	Cosim	BERT-F1	sec	tok
0a	Mistral-7B-Instruct	Baseline LLM only	no_think	noSF	7.60	80.64	62.22	5.75	471
	Mistral-7B-Instruct	Baseline LLM only	no_think	SF	7.11	80.59	62.61	10.37	968
	Mistral-7B-Instruct	Baseline LLM only	no_think	Δ	-0.49	-0.05	0.40	4.62	497
0b	Qwen3.5-9B	Baseline LLM only	no_think	noSF	7.63	82.06	63.39	11.84	582
	Qwen3.5-9B	Baseline LLM only	no_think	SF	7.83	81.83	64.09	19.21	1068
	Qwen3.5-9B	Baseline LLM only	no_think	Δ	+0.20	−0.23	+0.70	+7.37	+486
	Qwen3.5-9B	Baseline LLM only	think	noSF	0.54	77.04	25.99	29.34	1195
	Qwen3.5-9B	Baseline LLM only	think	SF	1.17	79.57	40.86	44.31	1938
	Qwen3.5-9B	Baseline LLM only	think	Δ	+0.63	+2.53	+14.87	+14.97	+743
1a	Mistral-7B-Instruct	e5 top 1	no_think	noSF	16.25	83.73	67.59	4.61	596
	Mistral-7B-Instruct	e5 top 1	no_think	SF	14.88	83.47	68.35	7.26	1187
	Mistral-7B-Instruct	e5 top 1	no_think	Δ	-1.37	-0.27	0.77	2.66	591
1b	Qwen3.5-9B	e5 top 1	no_think	noSF	19.38	85.37	72.51	2.00	379
	Qwen3.5-9B	e5 top 1	no_think	SF	18.89	85.03	72.17	4.01	820
	Qwen3.5-9B	e5 top 1	no_think	Δ	−0.49	−0.34	−0.34	+2.01	+441
	Qwen3.5-9B	e5 top 1	think	noSF	5.39	82.62	62.68	29.45	1316
	Qwen3.5-9B	e5 top 1	think	SF	2.28	78.68	40.74	44.36	2202
	Qwen3.5-9B	e5 top 1	think	Δ	−3.11	−3.94	−21.94	+14.91	+886

Continued on next page

Table 14 (continued)

#	Model	Experiment	Think	SF	Quality ↑			Cost ↓	
					ROUGE-L	Cosim	BERT-F1	sec	tok
2a	Mistral-7B-Instruct	e5 top 3	no_think	noSF	21.23	85.78	70.67	4.01	999
	Mistral-7B-Instruct	e5 top 3	no_think	SF	19.55	85.37	71.50	6.64	2018
	Mistral-7B-Instruct	e5 top 3	no_think	Δ	-1.68	-0.41	0.83	2.63	1019
2b	Qwen3.5-9B	e5 top 3	no_think	noSF	25.06	86.57	74.35	2.12	707
	Qwen3.5-9B	e5 top 3	no_think	SF	24.98	86.46	74.33	4.23	1476
	Qwen3.5-9B	e5 top 3	no_think	Δ	−0.08	−0.11	−0.02	+2.11	+769
	Qwen3.5-9B	e5 top 3	think	noSF	4.49	82.26	61.61	29.42	1642
	Qwen3.5-9B	e5 top 3	think	SF	1.56	78.39	36.48	44.50	2852
	Qwen3.5-9B	e5 top 3	think	Δ	−2.93	−3.87	−25.13	+15.08	+1210
3a	Mistral-7B-Instruct	e5 top 5	no_think	noSF	25.87	86.85	72.86	4.69	1549
	Mistral-7B-Instruct	e5 top 5	no_think	SF	22.56	85.92	72.60	7.71	3106
	Mistral-7B-Instruct	e5 top 5	no_think	Δ	-3.32	-0.93	-0.26	3.02	1557
3b	Qwen3.5-9B	e5 top 5	no_think	noSF	30.62	87.71	76.49	2.27	1110
	Qwen3.5-9B	e5 top 5	no_think	SF	29.79	87.68	76.28	4.56	2282
	Qwen3.5-9B	e5 top 5	no_think	Δ	−0.83	−0.03	−0.21	+2.29	+1172
	Qwen3.5-9B	e5 top 5	think	noSF	5.06	82.47	62.05	29.60	2060
	Qwen3.5-9B	e5 top 5	think	SF	1.47	78.22	33.96	44.70	3670
	Qwen3.5-9B	e5 top 5	think	Δ	−3.59	−4.25	−28.09	+15.10	+1610
4a	Mistral-7B-Instruct	rerank top 1	no_think	noSF	17.28	84.46	68.21	5.41	701
	Mistral-7B-Instruct	rerank top 1	no_think	SF	15.54	83.95	68.60	8.46	1396
	Mistral-7B-Instruct	rerank top 1	no_think	Δ	-1.74	-0.51	0.40	3.05	695
4b	Qwen3.5-9B	rerank top 1	no_think	noSF	19.55	85.32	71.42	3.31	481
	Qwen3.5-9B	rerank top 1	no_think	SF	18.78	85.14	71.25	6.26	1021
	Qwen3.5-9B	rerank top 1	no_think	Δ	−0.77	−0.18	−0.17	+2.95	+540
	Qwen3.5-9B	rerank top 1	think	noSF	5.14	82.31	62.00	29.69	1393
	Qwen3.5-9B	rerank top 1	think	SF	1.14	78.11	35.76	44.70	2346
	Qwen3.5-9B	rerank top 1	think	Δ	−4	−4.20	−26.24	+15.01	+953
5a	Mistral-7B-Instruct	rerank top 3	no_think	noSF	28.00	87.14	72.70	5.03	1179
	Mistral-7B-Instruct	rerank top 3	no_think	SF	26.02	86.48	73.43	8.00	2366
	Mistral-7B-Instruct	rerank top 3	no_think	Δ	-1.98	-0.66	0.74	2.97	1187
5b	Qwen3.5-9B	rerank top 3	no_think	noSF	30.44	87.91	75.72	2.96	843
	Qwen3.5-9B	rerank top 3	no_think	SF	28.05	87.35	75.12	5.38	1738
	Qwen3.5-9B	rerank top 3	no_think	Δ	−2.39	−0.56	−0.60	+2.42	+895
	Qwen3.5-9B	rerank top 3	think	noSF	4.60	82.41	62.21	29.84	1769
	Qwen3.5-9B	rerank top 3	think	SF	1.23	77.87	33.87	44.88	3097
	Qwen3.5-9B	rerank top 3	think	Δ	−3.37	−4.54	−28.34	+15.04	+1328
6a	Mistral-7B-Instruct	rerank top 5	no_think	noSF	32.10	88.19	75.33	4.69	1700
	Mistral-7B-Instruct	rerank top 5	no_think	SF	29.22	87.56	74.86	7.50	3417
	Mistral-7B-Instruct	rerank top 5	no_think	Δ	-2.88	-0.63	-0.48	2.81	1716
6b	Qwen3.5-9B	rerank top 5	no_think	noSF	37.14	89.40	78.35	3.28	1262
	Qwen3.5-9B	rerank top 5	no_think	SF	37.48	89.18	78.30	6.16	2580
	Qwen3.5-9B	rerank top 5	no_think	Δ	+0.34	−0.22	−0.05	+2.88	+1318
	Qwen3.5-9B	rerank top 5	think	noSF	5.23	82.55	62.14	29.87	2179
	Qwen3.5-9B	rerank top 5	think	SF	1.41	77.91	33.14	45.12	3917
	Qwen3.5-9B	rerank top 5	think	Δ	−3.82	−4.64	−29	+15.25	+1738
7a	Mistral-7B-Instruct	3-shot, no RAG	no_think	noSF	11.33	83.41	67.82	4.82	1348
	Mistral-7B-Instruct	3-shot, no RAG	no_think	SF	9.63	81.88	65.22	9.18	1834
	Mistral-7B-Instruct	3-shot, no RAG	no_think	Δ	-1.70	-1.52	-2.60	4.36	486
7b	Qwen3.5-9B	3-shot, no RAG	no_think	noSF	14.58	84.89	69.87	6.23	1085
	Qwen3.5-9B	3-shot, no RAG	no_think	SF	10.92	82.99	66.36	10.98	1480
	Qwen3.5-9B	3-shot, no RAG	no_think	Δ	−3.66	−1.90	−3.51	+4.75	+395
	Qwen3.5-9B	3-shot, no RAG	think	noSF	0.28	76.97	25.34	29.57	1898
	Qwen3.5-9B	3-shot, no RAG	think	SF	0.98	79.57	41.75	44.45	2641
	Qwen3.5-9B	3-shot, no RAG	think	Δ	+0.70	+2.60	+16.41	+14.88	+743
8a	Mistral-7B-Instruct	3-shot + rerank top 5	no_think	noSF	35.53	88.94	76.96	4.02	2590

Continued on next page

Table 14 (continued)

#	Model	Experiment	Think	SF	Quality ↑			Cost ↓	
					ROUGE-L	Cosim	BERT-F1	sec	tok
	Mistral-7B-Instruct	3-shot + rerank top 5	no_think	SF	31.43	87.90	75.54	7.12	4319
	Mistral-7B-Instruct	3-shot + rerank top 5	no_think	Δ	-4.11	-1.04	-1.42	3.10	1729
⇒ 8b	Qwen3.5-9B	3-shot + rerank top 5	no_think	noSF	38.68	89.61	78.91	4.10	1989
	Qwen3.5-9B	3-shot + rerank top 5	no_think	SF	38.52	89.34	78.80	6.79	3301
	Qwen3.5-9B	3-shot + rerank top 5	no_think	Δ	−0.16	−0.27	−0.11	+2.69	+1312
	Qwen3.5-9B	3-shot + rerank top 5	think	noSF	6.32	82.71	62.13	30.17	2873
	Qwen3.5-9B	3-shot + rerank top 5	think	SF	2.97	78.70	37.28	45.39	4611
	Qwen3.5-9B	3-shot + rerank top 5	think	Δ	−3.35	−4.01	−24.85	+15.22	+1738
9a	Mistral-7B-Instruct	cross-domain retrieval	no_think	noSF	8.18	81.27	63.51	7.65	2981
	Mistral-7B-Instruct	cross-domain retrieval	no_think	SF	8.18	81.30	64.14	12.66	5949
	Mistral-7B-Instruct	cross-domain retrieval	no_think	Δ	0.00	0.03	0.63	5.00	2967
9b	Qwen3.5-9B	cross-domain retrieval	no_think	noSF	8.20	82.81	65.12	4.27	2227
	Qwen3.5-9B	cross-domain retrieval	no_think	SF	8.26	82.82	65.21	8.35	4516
	Qwen3.5-9B	cross-domain retrieval	no_think	Δ	+0.06	+0.01	+0.09	+4.08	+2289
	Qwen3.5-9B	cross-domain retrieval	think	noSF	2.09	81.45	59.66	30.15	3124
	Qwen3.5-9B	cross-domain retrieval	think	SF	0.37	76.83	27.47	45.61	5794
	Qwen3.5-9B	cross-domain retrieval	think	Δ	−1.72	−4.62	−32.19	+15.46	+2670
10a	Mistral-7B-Instruct	mixed-domain retrieval	no_think	noSF	32.10	88.19	75.33	4.68	1700
	Mistral-7B-Instruct	mixed-domain retrieval	no_think	SF	29.22	87.56	74.86	7.49	3417
	Mistral-7B-Instruct	mixed-domain retrieval	no_think	Δ	-2.88	-0.63	-0.48	2.81	1716
10b	Qwen3.5-9B	mixed-domain retrieval	no_think	noSF	37.14	89.40	78.35	3.28	1262
	Qwen3.5-9B	mixed-domain retrieval	no_think	SF	37.48	89.18	78.30	6.15	2580
	Qwen3.5-9B	mixed-domain retrieval	no_think	Δ	+0.34	−0.22	−0.05	+2.87	+1318
	Qwen3.5-9B	mixed-domain retrieval	think	noSF	5.23	82.55	62.14	29.98	2179
	Qwen3.5-9B	mixed-domain retrieval	think	SF	1.41	77.91	33.14	45.17	3917
	Qwen3.5-9B	mixed-domain retrieval	think	Δ	−3.82	−4.64	−29	+15.19	+1738

Table 15 covers the CasiMédicos-Arg dev set (multiple-choice, 55 examples). Because answers are short option labels, BERT-F1 and Cosim are more diagnostic than ROUGE-L here.

Table 15: Spanish CasiMedicos dev ablation (55 examples, multiple-choice). Same metrics as Table 14.

#	Model	Experiment	Think	SF	Quality ↑			Cost ↓	
					ROUGE-L	Cosim	BERT-F1	sec	tok
0a	Mistral-7B-Instruct	Baseline LLM only	no_think	noSF	31.89	90.10	76.84	4.86	626
	Mistral-7B-Instruct	Baseline LLM only	no_think	SF	32.56	90.24	77.09	8.83	1288
	Mistral-7B-Instruct	Baseline LLM only	no_think	Δ	0.67	0.14	0.25	3.97	662
0b	Qwen3.5-9B	Baseline LLM only	no_think	noSF	41.98	92.16	79.27	10.86	704
	Qwen3.5-9B	Baseline LLM only	no_think	SF	43.11	92.06	79.87	16.90	1299
	Qwen3.5-9B	Baseline LLM only	no_think	Δ	+1.13	−0.10	+0.60	+6.04	+595
	Qwen3.5-9B	Baseline LLM only	think	noSF	0.80	79.21	29.23	29.46	1351
	Qwen3.5-9B	Baseline LLM only	think	SF	1.22	81.13	47.95	44.45	2250
	Qwen3.5-9B	Baseline LLM only	think	Δ	+0.42	+1.92	+18.72	+14.99	+899
1a	Mistral-7B-Instruct	e5 top 1	no_think	noSF	33.36	90.31	76.72	4.48	974
	Mistral-7B-Instruct	e5 top 1	no_think	SF	31.15	89.78	76.57	7.48	1965
	Mistral-7B-Instruct	e5 top 1	no_think	Δ	-2.21	-0.53	-0.15	3.00	990
1b	Qwen3.5-9B	e5 top 1	no_think	noSF	47.64	91.91	80.94	3.79	750
	Qwen3.5-9B	e5 top 1	no_think	SF	47.60	92.06	81.07	6.98	1541
	Qwen3.5-9B	e5 top 1	no_think	Δ	−0.04	+0.15	+0.13	+3.19	+791
	Qwen3.5-9B	e5 top 1	think	noSF	4.86	82.64	60.55	29.32	1627
	Qwen3.5-9B	e5 top 1	think	SF	1.35	79.09	34.23	44.38	2822
	Qwen3.5-9B	e5 top 1	think	Δ	−3.51	−3.55	−26.32	+15.06	+1195
2a	Mistral-7B-Instruct	e5 top 3	no_think	noSF	40.39	91.69	79.51	4.28	1744

Continued on next page

Table 15 (continued)

#	Model	Experiment	Think	SF	Quality ↑			Cost ↓	
					ROUGE-L	Cosim	BERT-F1	sec	tok
	Mistral-7B-Instruct	e5 top 3	no_think	SF	40.10	91.35	79.19	8.04	3545
	Mistral-7B-Instruct	e5 top 3	no_think	Δ	-0.29	-0.33	-0.32	3.76	1801
2b	Qwen3.5-9B	e5 top 3	no_think	noSF	51.74	93.24	82.75	4.22	1370
	Qwen3.5-9B	e5 top 3	no_think	SF	50.08	92.73	82.35	7.96	2785
	Qwen3.5-9B	e5 top 3	no_think	Δ	−1.66	−0.51	−0.40	+3.74	+1415
	Qwen3.5-9B	e5 top 3	think	noSF	4.33	82.55	59.38	29.66	2253
	Qwen3.5-9B	e5 top 3	think	SF	0.96	78.78	32.21	44.81	4056
	Qwen3.5-9B	e5 top 3	think	Δ	−3.37	−3.77	−27.17	+15.15	+1803
3a	Mistral-7B-Instruct	e5 top 5	no_think	noSF	39.62	91.46	79.00	4.18	2477
	Mistral-7B-Instruct	e5 top 5	no_think	SF	39.42	91.32	79.36	7.72	5006
	Mistral-7B-Instruct	e5 top 5	no_think	Δ	-0.19	-0.14	0.37	3.54	2529
3b	Qwen3.5-9B	e5 top 5	no_think	noSF	49.80	92.89	82.12	4.58	1955
	Qwen3.5-9B	e5 top 5	no_think	SF	50.73	93.03	82.66	8.51	3948
	Qwen3.5-9B	e5 top 5	no_think	Δ	+0.93	+0.14	+0.54	+3.93	+1993
	Qwen3.5-9B	e5 top 5	think	noSF	3.42	81.93	53.17	29.76	2829
	Qwen3.5-9B	e5 top 5	think	SF	2.24	78.80	30.87	45.13	5208
	Qwen3.5-9B	e5 top 5	think	Δ	−1.18	−3.13	−22.30	+15.37	+2379
4a	Mistral-7B-Instruct	rerank top 1	no_think	noSF	31.59	89.98	76.80	4.92	917
	Mistral-7B-Instruct	rerank top 1	no_think	SF	31.99	89.89	77.16	8.11	1858
	Mistral-7B-Instruct	rerank top 1	no_think	Δ	0.41	-0.09	0.37	3.19	940
4b	Qwen3.5-9B	rerank top 1	no_think	noSF	41.63	90.68	79.05	4.71	725
	Qwen3.5-9B	rerank top 1	no_think	SF	42.46	90.82	79.80	8.25	1478
	Qwen3.5-9B	rerank top 1	no_think	Δ	+0.83	+0.14	+0.75	+3.54	+753
	Qwen3.5-9B	rerank top 1	think	noSF	2.91	82.13	59.95	29.82	1598
	Qwen3.5-9B	rerank top 1	think	SF	0.97	78.74	33.02	44.87	2747
	Qwen3.5-9B	rerank top 1	think	Δ	−1.94	−3.39	−26.93	+15.05	+1149
5a	Mistral-7B-Instruct	rerank top 3	no_think	noSF	38.75	91.07	78.82	4.16	1586
	Mistral-7B-Instruct	rerank top 3	no_think	SF	37.89	90.83	78.56	7.73	3244
	Mistral-7B-Instruct	rerank top 3	no_think	Δ	-0.86	-0.25	-0.26	3.57	1659
5b	Qwen3.5-9B	rerank top 3	no_think	noSF	46.63	91.74	80.62	4.95	1279
	Qwen3.5-9B	rerank top 3	no_think	SF	45.69	91.50	80.34	8.91	2591
	Qwen3.5-9B	rerank top 3	no_think	Δ	−0.94	−0.24	−0.28	+3.96	+1312
	Qwen3.5-9B	rerank top 3	think	noSF	3.99	82.23	58.48	29.93	2146
	Qwen3.5-9B	rerank top 3	think	SF	1.29	78.98	34.87	45.19	3842
	Qwen3.5-9B	rerank top 3	think	Δ	−2.70	−3.25	−23.61	+15.26	+1696
6a	Mistral-7B-Instruct	rerank top 5	no_think	noSF	38.01	91.04	77.80	4.59	2412
	Mistral-7B-Instruct	rerank top 5	no_think	SF	39.95	91.41	79.77	8.55	4896
	Mistral-7B-Instruct	rerank top 5	no_think	Δ	1.95	0.37	1.97	3.97	2484
6b	Qwen3.5-9B	rerank top 5	no_think	noSF	48.27	91.93	81.27	5.15	1915
	Qwen3.5-9B	rerank top 5	no_think	SF	48.02	91.93	81.65	9.10	3856
	Qwen3.5-9B	rerank top 5	no_think	Δ	−0.25	+0.00	+0.38	+3.95	+1941
	Qwen3.5-9B	rerank top 5	think	noSF	3.99	82.07	56.34	30.14	2779
	Qwen3.5-9B	rerank top 5	think	SF	1.40	78.19	27.66	45.44	5108
	Qwen3.5-9B	rerank top 5	think	Δ	−2.59	−3.88	−28.68	+15.30	+2329
7a	Mistral-7B-Instruct	3-shot, no RAG	no_think	noSF	34.12	90.46	77.89	4.30	1973
	Mistral-7B-Instruct	3-shot, no RAG	no_think	SF	34.04	90.60	78.03	8.06	2627
	Mistral-7B-Instruct	3-shot, no RAG	no_think	Δ	-0.09	0.13	0.14	3.76	654
7b	Qwen3.5-9B	3-shot, no RAG	no_think	noSF	50.34	93.70	82.92	8.79	1707
⇒	Qwen3.5-9B	3-shot, no RAG	no_think	SF	51.55	93.71	83.28	14.36	2286
	Qwen3.5-9B	3-shot, no RAG	no_think	Δ	+1.21	+0.01	+0.36	+5.57	+579
	Qwen3.5-9B	3-shot, no RAG	think	noSF	2.23	79.55	31.06	29.67	2432
	Qwen3.5-9B	3-shot, no RAG	think	SF	1.06	81.10	48.13	44.70	3331
	Qwen3.5-9B	3-shot, no RAG	think	Δ	−1.17	+1.55	+17.07	+15.03	+899
8a	Mistral-7B-Instruct	3-shot + rerank top 5	no_think	noSF	41.17	91.79	80.20	4.87	3795
	Mistral-7B-Instruct	3-shot + rerank top 5	no_think	SF	42.39	91.85	80.66	8.79	6276

Continued on next page

Table 15 (continued)

#	Model	Experiment	Think	SF	Quality ↑			Cost ↓	
					ROUGE-L	Cosim	BERT-F1	sec	tok
	Mistral-7B-Instruct	3-shot + rerank top 5	no_think	Δ	1.21	0.07	0.46	3.92	2481
8b	Qwen3.5-9B	3-shot + rerank top 5	no_think	noSF	51.14	92.66	82.39	5.37	2996
	Qwen3.5-9B	3-shot + rerank top 5	no_think	SF	49.15	92.42	82.09	8.87	4926
	Qwen3.5-9B	3-shot + rerank top 5	no_think	Δ	−1.99	−0.24	−0.30	+3.50	+1930
	Qwen3.5-9B	3-shot + rerank top 5	think	noSF	3.05	80.73	42.28	30.36	3860
	Qwen3.5-9B	3-shot + rerank top 5	think	SF	2.79	79.07	33.17	45.74	6190
	Qwen3.5-9B	3-shot + rerank top 5	think	Δ	−0.26	−1.66	−9.11	+15.38	+2330
9a	Mistral-7B-Instruct	cross-domain retrieval	no_think	noSF	33.72	89.97	76.27	6.45	5243
	Mistral-7B-Instruct	cross-domain retrieval	no_think	SF	31.67	89.50	76.52	10.69	10498
	Mistral-7B-Instruct	cross-domain retrieval	no_think	Δ	-2.05	-0.47	0.25	4.24	5255
9b	Qwen3.5-9B	cross-domain retrieval	no_think	noSF	22.99	85.44	71.50	6.01	4068
	Qwen3.5-9B	cross-domain retrieval	no_think	SF	17.06	84.08	69.31	10.92	8171
	Qwen3.5-9B	cross-domain retrieval	no_think	Δ	−5.93	−1.36	−2.19	+4.91	+4103
	Qwen3.5-9B	cross-domain retrieval	think	noSF	1.93	81.78	58.91	30.87	4920
	Qwen3.5-9B	cross-domain retrieval	think	SF	0.64	78.23	29.51	46.67	9391
	Qwen3.5-9B	cross-domain retrieval	think	Δ	−1.29	−3.55	−29.40	+15.80	+4471
10a	Mistral-7B-Instruct	mixed-domain retrieval	no_think	noSF	38.01	91.04	77.80	4.66	2412
	Mistral-7B-Instruct	mixed-domain retrieval	no_think	SF	39.95	91.41	79.77	8.63	4896
	Mistral-7B-Instruct	mixed-domain retrieval	no_think	Δ	1.95	0.37	1.97	3.97	2484
10b	Qwen3.5-9B	mixed-domain retrieval	no_think	noSF	48.27	91.93	81.27	5.25	1915
	Qwen3.5-9B	mixed-domain retrieval	no_think	SF	48.02	91.93	81.65	9.09	3856
	Qwen3.5-9B	mixed-domain retrieval	no_think	Δ	−0.25	+0.00	+0.38	+3.84	+1941
	Qwen3.5-9B	mixed-domain retrieval	think	noSF	3.99	82.07	56.34	30.17	2779
	Qwen3.5-9B	mixed-domain retrieval	think	SF	1.40	78.19	27.66	45.51	5108
	Qwen3.5-9B	mixed-domain retrieval	think	Δ	−2.59	−3.88	−28.68	+15.34	+2329

Table 16 covers the mixed dev set (SNS1064 + CasiMédicos-Arg combined, 161 examples), providing a single aggregate view across both task types.

Table 16: Spanish SNS1064+CasiMedicos dev ablation (161 examples, mixed). Same metrics as Table 14.

#	Model	Experiment	Think	SF	Quality ↑			Cost ↓	
					ROUGE-L	Cosim	BERT-F1	sec	tok
0a	Mistral-7B-Instruct	Baseline LLM only	no_think	noSF	15.89	83.87	67.21	5.44	524
	Mistral-7B-Instruct	Baseline LLM only	no_think	SF	15.80	83.89	67.56	9.83	1077
	Mistral-7B-Instruct	Baseline LLM only	no_think	Δ	-0.09	0.02	0.35	4.39	553
0b	Qwen3.5-9B	Baseline LLM only	no_think	noSF	19.36	85.51	68.82	11.49	623
	Qwen3.5-9B	Baseline LLM only	no_think	SF	19.88	85.32	69.48	18.42	1147
	Qwen3.5-9B	Baseline LLM only	no_think	Δ	+0.52	−0.19	+0.66	+6.93	+524
	Qwen3.5-9B	Baseline LLM only	think	noSF	0.63	77.78	27.10	29.38	1248
	Qwen3.5-9B	Baseline LLM only	think	SF	1.19	80.11	43.28	44.38	2044
	Qwen3.5-9B	Baseline LLM only	think	Δ	+0.56	+2.33	+16.18	+15.00	+796
1a	Mistral-7B-Instruct	e5 top 1	no_think	noSF	22.09	85.98	70.71	4.56	725
	Mistral-7B-Instruct	e5 top 1	no_think	SF	20.44	85.63	71.16	7.34	1453
	Mistral-7B-Instruct	e5 top 1	no_think	Δ	-1.65	-0.36	0.45	2.78	727
1b	Qwen3.5-9B	e5 top 1	no_think	noSF	29.04	87.61	75.39	2.61	505
	Qwen3.5-9B	e5 top 1	no_think	SF	28.70	87.44	75.21	5.04	1066
	Qwen3.5-9B	e5 top 1	no_think	Δ	−0.34	−0.17	−0.18	+2.43	+561
	Qwen3.5-9B	e5 top 1	think	noSF	5.21	82.63	61.96	29.35	1422
	Qwen3.5-9B	e5 top 1	think	SF	1.96	78.82	38.52	44.46	2413
	Qwen3.5-9B	e5 top 1	think	Δ	−3.25	−3.81	−23.44	+15.11	+991
2a	Mistral-7B-Instruct	e5 top 3	no_think	noSF	27.77	87.80	73.69	4.10	1254
	Mistral-7B-Instruct	e5 top 3	no_think	SF	26.57	87.42	74.12	7.12	2540
	Mistral-7B-Instruct	e5 top 3	no_think	Δ	-1.20	-0.38	0.44	3.02	1286

Continued on next page

Table 16 (continued)

#	Model	Experiment	Think	SF	Quality ↑			Cost ↓	
					ROUGE-L	Cosim	BERT-F1	sec	tok
2b	Qwen3.5-9B	e5 top 3	no_think	noSF	34.17	88.85	77.22	2.84	933
	Qwen3.5-9B	e5 top 3	no_think	SF	33.56	88.60	77.07	5.52	1923
	Qwen3.5-9B	e5 top 3	no_think	Δ	−0.61	−0.25	−0.15	+2.68	+990
	Qwen3.5-9B	e5 top 3	think	noSF	4.44	82.36	60.84	29.56	1850
	Qwen3.5-9B	e5 top 3	think	SF	1.36	78.52	35.02	44.79	3263
	Qwen3.5-9B	e5 top 3	think	Δ	−3.08	−3.84	−25.82	+15.23	+1413
3a	Mistral-7B-Instruct	e5 top 5	no_think	noSF	30.57	88.42	74.95	4.51	1866
	Mistral-7B-Instruct	e5 top 5	no_think	SF	28.32	87.76	74.91	7.71	3755
	Mistral-7B-Instruct	e5 top 5	no_think	Δ	-2.25	-0.66	-0.04	3.20	1889
3b	Qwen3.5-9B	e5 top 5	no_think	noSF	37.17	89.48	78.41	3.07	1399
	Qwen3.5-9B	e5 top 5	no_think	SF	36.94	89.51	78.46	5.90	2851
	Qwen3.5-9B	e5 top 5	no_think	Δ	−0.23	+0.03	+0.05	+2.83	+1452
	Qwen3.5-9B	e5 top 5	think	noSF	4.50	82.28	59.02	29.68	2322
	Qwen3.5-9B	e5 top 5	think	SF	1.73	78.41	32.90	44.95	4195
	Qwen3.5-9B	e5 top 5	think	Δ	−2.77	−3.87	−26.12	+15.27	+1873
4a	Mistral-7B-Instruct	rerank top 1	no_think	noSF	22.17	86.35	71.14	5.28	775
	Mistral-7B-Instruct	rerank top 1	no_think	SF	21.16	85.98	71.53	8.37	1554
	Mistral-7B-Instruct	rerank top 1	no_think	Δ	-1.01	-0.37	0.39	3.10	779
4b	Qwen3.5-9B	rerank top 1	no_think	noSF	27.09	87.15	74.03	3.86	565
	Qwen3.5-9B	rerank top 1	no_think	SF	26.87	87.08	74.17	7.01	1177
	Qwen3.5-9B	rerank top 1	no_think	Δ	−0.22	−0.07	+0.14	+3.15	+612
	Qwen3.5-9B	rerank top 1	think	noSF	4.38	82.25	61.30	29.77	1463
	Qwen3.5-9B	rerank top 1	think	SF	1.08	78.32	34.82	44.83	2483
	Qwen3.5-9B	rerank top 1	think	Δ	−3.30	−3.93	−26.48	+15.06	+1020
5a	Mistral-7B-Instruct	rerank top 3	no_think	noSF	31.67	88.48	74.79	4.75	1318
	Mistral-7B-Instruct	rerank top 3	no_think	SF	30.08	87.96	75.18	7.92	2666
	Mistral-7B-Instruct	rerank top 3	no_think	Δ	-1.60	-0.52	0.40	3.18	1348
5b	Qwen3.5-9B	rerank top 3	no_think	noSF	35.97	89.22	77.39	3.69	992
	Qwen3.5-9B	rerank top 3	no_think	SF	34.08	88.77	76.90	6.61	2030
	Qwen3.5-9B	rerank top 3	no_think	Δ	−1.89	−0.45	−0.49	+2.92	+1038
	Qwen3.5-9B	rerank top 3	think	noSF	4.40	82.35	60.94	29.91	1898
	Qwen3.5-9B	rerank top 3	think	SF	1.25	78.25	34.22	45.04	3352
	Qwen3.5-9B	rerank top 3	think	Δ	−3.15	−4.10	−26.72	+15.13	+1454
6a	Mistral-7B-Instruct	rerank top 5	no_think	noSF	34.12	89.16	76.17	4.64	1944
	Mistral-7B-Instruct	rerank top 5	no_think	SF	32.89	88.88	76.53	7.85	3922
	Mistral-7B-Instruct	rerank top 5	no_think	Δ	-1.23	-0.29	0.36	3.21	1978
6b	Qwen3.5-9B	rerank top 5	no_think	noSF	40.95	90.26	79.35	3.94	1485
⇒	Qwen3.5-9B	rerank top 5	no_think	SF	41.08	90.12	79.44	7.07	3016
	Qwen3.5-9B	rerank top 5	no_think	Δ	+0.13	−0.14	+0.09	+3.13	+1531
	Qwen3.5-9B	rerank top 5	think	noSF	4.81	82.39	60.16	30.00	2384
	Qwen3.5-9B	rerank top 5	think	SF	1.40	78.00	31.27	45.23	4324
	Qwen3.5-9B	rerank top 5	think	Δ	−3.41	−4.39	−28.89	+15.23	+1940
7a	Mistral-7B-Instruct	3-shot, no RAG	no_think	noSF	18.94	85.20	70.46	4.96	1682
	Mistral-7B-Instruct	3-shot, no RAG	no_think	SF	17.92	84.44	69.18	9.14	2226
	Mistral-7B-Instruct	3-shot, no RAG	no_think	Δ	-1.02	-0.77	-1.27	4.17	544
7b	Qwen3.5-9B	3-shot, no RAG	no_think	noSF	23.32	87.22	72.93	7.44	1394
	Qwen3.5-9B	3-shot, no RAG	no_think	SF	22.84	86.33	71.71	12.71	1859
	Qwen3.5-9B	3-shot, no RAG	no_think	Δ	−0.48	−0.89	−1.22	+5.27	+465
	Qwen3.5-9B	3-shot, no RAG	think	noSF	0.46	77.83	26.93	29.65	2165
	Qwen3.5-9B	3-shot, no RAG	think	SF	1.46	80.26	44.88	44.58	2961
	Qwen3.5-9B	3-shot, no RAG	think	Δ	+1	+2.43	+17.95	+14.93	+796
8a	Mistral-7B-Instruct	3-shot + rerank top 5	no_think	noSF	36.38	89.38	76.87	4.24	3105
	Mistral-7B-Instruct	3-shot + rerank top 5	no_think	SF	34.22	88.88	76.73	7.63	5091
	Mistral-7B-Instruct	3-shot + rerank top 5	no_think	Δ	-2.16	-0.51	-0.14	3.40	1987
8b	Qwen3.5-9B	3-shot + rerank top 5	no_think	noSF	41.04	90.19	79.32	4.67	2422

Continued on next page

Table 16 (continued)

#	Model	Experiment	Think	SF	Quality ↑			Cost ↓	
					ROUGE-L	Cosim	BERT-F1	sec	tok
	Qwen3.5-9B	3-shot + rerank top 5	no_think	SF	40.90	90.07	79.17	7.81	3950
	Qwen3.5-9B	3-shot + rerank top 5	no_think	Δ	−0.14	−0.12	−0.15	+3.14	+1528
	Qwen3.5-9B	3-shot + rerank top 5	think	noSF	4.67	81.93	56.19	30.24	3301
	Qwen3.5-9B	3-shot + rerank top 5	think	SF	1.49	78.14	32.02	45.44	5241
	Qwen3.5-9B	3-shot + rerank top 5	think	Δ	−3.18	−3.79	−24.17	+15.20	+1940
9a	Mistral-7B-Instruct	cross-domain retrieval	no_think	noSF	32.65	88.80	75.65	5.31	2911
	Mistral-7B-Instruct	cross-domain retrieval	no_think	SF	30.06	88.22	75.42	8.61	5836
	Mistral-7B-Instruct	cross-domain retrieval	no_think	Δ	-2.60	-0.57	-0.23	3.30	2925
9b	Qwen3.5-9B	cross-domain retrieval	no_think	noSF	32.31	88.05	76.01	4.19	2221
	Qwen3.5-9B	cross-domain retrieval	no_think	SF	30.51	87.44	75.23	7.76	4490
	Qwen3.5-9B	cross-domain retrieval	no_think	Δ	−1.80	−0.61	−0.78	+3.57	+2269
	Qwen3.5-9B	cross-domain retrieval	think	noSF	4.10	82.29	61.03	30.24	3116
	Qwen3.5-9B	cross-domain retrieval	think	SF	1.14	78.02	31.90	45.67	5787
	Qwen3.5-9B	cross-domain retrieval	think	Δ	−2.96	−4.27	−29.13	+15.43	+2671
10a	Mistral-7B-Instruct	mixed-domain retrieval	no_think	noSF	18.37	84.61	68.39	6.52	2787
	Mistral-7B-Instruct	mixed-domain retrieval	no_think	SF	19.04	84.75	69.48	11.17	5589
	Mistral-7B-Instruct	mixed-domain retrieval	no_think	Δ	0.67	0.15	1.09	4.65	2802
10b	Qwen3.5-9B	mixed-domain retrieval	no_think	noSF	21.89	85.93	70.64	4.61	2120
	Qwen3.5-9B	mixed-domain retrieval	no_think	SF	21.84	85.93	70.82	8.66	4290
	Qwen3.5-9B	mixed-domain retrieval	no_think	Δ	−0.05	+0.00	+0.18	+4.05	+2170
	Qwen3.5-9B	mixed-domain retrieval	think	noSF	2.74	81.66	58.52	30.22	3006
	Qwen3.5-9B	mixed-domain retrieval	think	SF	0.72	77.29	27.54	45.63	5560
	Qwen3.5-9B	mixed-domain retrieval	think	Δ	−2.02	−4.37	−30.98	+15.41	+2554

6.2 Basque results

For Basque, row labels follow the same Xa/Xb pattern but a = Llama-3.1-8B-Instruct and b = Latxa-Llama-3.1-8B-Instruct (the Basque-adapted variant); the SF column shows noSF/SF/Δ for both models.

Table 17 covers the Basque SNS1064 EU dev set (open-answer, 106 examples).

Table 17: Basque SNS1064 EU dev ablation (106 examples, open-answer). Same metrics as Table 14.

#	Model	Experiment	SF	Quality ↑			Cost ↓	
				ROUGE-L	Cosim	BERT-F1	sec	tok
0a	Llama-3.1-8B-Instruct	Baseline LLM only	noSF	4.67	80.43	60.01	6.12	509
	Llama-3.1-8B-Instruct	Baseline LLM only	SF	4.69	80.59	62.25	11.99	1084
	Llama-3.1-8B-Instruct	Baseline LLM only	Δ	0.02	0.16	2.23	5.87	575
0b	Latxa-Llama-3.1-8B-Instruct	Baseline LLM only	noSF	5.66	80.75	62.66	6.05	506
	Latxa-Llama-3.1-8B-Instruct	Baseline LLM only	SF	5.41	80.66	62.61	11.85	1078
	Latxa-Llama-3.1-8B-Instruct	Baseline LLM only	Δ	-0.25	-0.09	-0.04	5.80	572
1a	Llama-3.1-8B-Instruct	e5 top 1	noSF	16.89	85.03	70.42	3.43	589
	Llama-3.1-8B-Instruct	e5 top 1	SF	14.50	84.56	69.29	7.14	1268
	Llama-3.1-8B-Instruct	e5 top 1	Δ	-2.39	-0.47	-1.14	3.71	679
1b	Latxa-Llama-3.1-8B-Instruct	e5 top 1	noSF	21.23	86.31	72.49	2.43	548
	Latxa-Llama-3.1-8B-Instruct	e5 top 1	SF	17.89	85.39	70.90	5.98	1221
	Latxa-Llama-3.1-8B-Instruct	e5 top 1	Δ	-3.34	-0.93	-1.60	3.55	673
2a	Llama-3.1-8B-Instruct	e5 top 3	noSF	24.25	86.85	73.65	2.82	938
	Llama-3.1-8B-Instruct	e5 top 3	SF	19.29	85.68	71.33	6.08	1972
	Llama-3.1-8B-Instruct	e5 top 3	Δ	-4.96	-1.18	-2.33	3.26	1034
2b	Latxa-Llama-3.1-8B-Instruct	e5 top 3	noSF	25.22	86.77	73.30	2.41	921
	Latxa-Llama-3.1-8B-Instruct	e5 top 3	SF	25.99	86.90	73.69	5.33	1941

Continued on next page

Table 17 (continued)

#	Model	Experiment	SF	Quality ↑			Cost ↓	
				ROUGE-L	Cosim	BERT-F1	sec	tok
	Latxa-Llama-3.1-8B-Instruct	e5 top 3	Δ	0.78	0.13	0.38	2.92	1020
3a	Llama-3.1-8B-Instruct	e5 top 5	noSF	29.39	87.53	75.13	2.40	1344
	Llama-3.1-8B-Instruct	e5 top 5	SF	24.89	86.69	73.15	5.44	2794
	Llama-3.1-8B-Instruct	e5 top 5	Δ	-4.50	-0.85	-1.98	3.04	1449
3b	Latxa-Llama-3.1-8B-Instruct	e5 top 5	noSF	33.61	88.67	76.35	2.63	1353
	Latxa-Llama-3.1-8B-Instruct	e5 top 5	SF	30.80	88.07	75.65	5.07	2778
	Latxa-Llama-3.1-8B-Instruct	e5 top 5	Δ	-2.80	-0.59	-0.70	2.44	1425
4a	Llama-3.1-8B-Instruct	rerank top 1	noSF	19.65	85.77	71.42	4.14	656
	Llama-3.1-8B-Instruct	rerank top 1	SF	15.20	84.64	69.25	8.58	1414
	Llama-3.1-8B-Instruct	rerank top 1	Δ	-4.45	-1.13	-2.17	4.44	758
4b	Latxa-Llama-3.1-8B-Instruct	rerank top 1	noSF	22.55	86.73	72.55	3.52	630
	Latxa-Llama-3.1-8B-Instruct	rerank top 1	SF	22.06	86.51	72.59	7.34	1362
	Latxa-Llama-3.1-8B-Instruct	rerank top 1	Δ	-0.49	-0.22	0.04	3.82	732
5a	Llama-3.1-8B-Instruct	rerank top 3	noSF	28.59	87.71	74.70	3.85	1092
	Llama-3.1-8B-Instruct	rerank top 3	SF	24.06	86.67	72.85	7.78	2276
	Llama-3.1-8B-Instruct	rerank top 3	Δ	-4.53	-1.04	-1.86	3.93	1184
5b	Latxa-Llama-3.1-8B-Instruct	rerank top 3	noSF	31.18	88.11	75.58	3.42	1074
	Latxa-Llama-3.1-8B-Instruct	rerank top 3	SF	30.22	87.95	75.33	6.67	2231
	Latxa-Llama-3.1-8B-Instruct	rerank top 3	Δ	-0.96	-0.16	-0.26	3.25	1157
6a	Llama-3.1-8B-Instruct	rerank top 5	noSF	32.97	88.45	76.33	3.29	1498
	Llama-3.1-8B-Instruct	rerank top 5	SF	30.57	87.74	75.18	6.54	3083
	Llama-3.1-8B-Instruct	rerank top 5	Δ	-2.40	-0.71	-1.14	3.24	1585
6b	Latxa-Llama-3.1-8B-Instruct	rerank top 5	noSF	34.14	88.75	76.50	3.22	1495
	Latxa-Llama-3.1-8B-Instruct	rerank top 5	SF	32.32	88.26	75.83	6.06	3063
	Latxa-Llama-3.1-8B-Instruct	rerank top 5	Δ	-1.82	-0.48	-0.67	2.84	1568
7a	Llama-3.1-8B-Instruct	3-shot, no RAG	noSF	9.10	83.45	66.28	4.43	1120
	Llama-3.1-8B-Instruct	3-shot, no RAG	SF	5.94	81.43	62.96	10.12	1687
	Llama-3.1-8B-Instruct	3-shot, no RAG	Δ	-3.16	-2.02	-3.31	5.69	568
7b	Latxa-Llama-3.1-8B-Instruct	3-shot, no RAG	noSF	10.45	83.78	66.66	3.69	1089
	Latxa-Llama-3.1-8B-Instruct	3-shot, no RAG	SF	7.12	82.12	65.15	8.01	1600
	Latxa-Llama-3.1-8B-Instruct	3-shot, no RAG	Δ	-3.32	-1.66	-1.51	4.31	511
8a	Llama-3.1-8B-Instruct	3-shot + rerank top 5	noSF	32.54	88.34	76.21	3.21	2176
	Llama-3.1-8B-Instruct	3-shot + rerank top 5	SF	29.63	87.82	74.98	6.57	3766
	Llama-3.1-8B-Instruct	3-shot + rerank top 5	Δ	-2.92	-0.52	-1.24	3.36	1589
⇒ 8b	Latxa-Llama-3.1-8B-Instruct	3-shot + rerank top 5	noSF	35.76	89.01	77.56	3.20	2175
	Latxa-Llama-3.1-8B-Instruct	3-shot + rerank top 5	SF	33.88	88.63	76.45	6.33	3755
	Latxa-Llama-3.1-8B-Instruct	3-shot + rerank top 5	Δ	-1.88	-0.38	-1.11	3.13	1580
9a	Llama-3.1-8B-Instruct	cross-domain retrieval	noSF	6.75	81.46	60.81	5.49	2273
	Llama-3.1-8B-Instruct	cross-domain retrieval	SF	6.77	81.58	63.36	10.51	4615
	Llama-3.1-8B-Instruct	cross-domain retrieval	Δ	0.03	0.12	2.55	5.02	2342
9b	Latxa-Llama-3.1-8B-Instruct	cross-domain retrieval	noSF	7.16	81.77	62.82	4.33	2225
	Latxa-Llama-3.1-8B-Instruct	cross-domain retrieval	SF	5.16	81.09	59.78	9.44	4571
	Latxa-Llama-3.1-8B-Instruct	cross-domain retrieval	Δ	-2.01	-0.68	-3.04	5.11	2346
10a	Llama-3.1-8B-Instruct	mixed-domain retrieval	noSF	32.97	88.45	76.33	3.31	1498
	Llama-3.1-8B-Instruct	mixed-domain retrieval	SF	30.57	87.74	75.18	6.55	3083
	Llama-3.1-8B-Instruct	mixed-domain retrieval	Δ	-2.40	-0.71	-1.14	3.24	1585
10b	Latxa-Llama-3.1-8B-Instruct	mixed-domain retrieval	noSF	34.14	88.75	76.50	3.23	1495
	Latxa-Llama-3.1-8B-Instruct	mixed-domain retrieval	SF	32.32	88.26	75.83	6.07	3063
	Latxa-Llama-3.1-8B-Instruct	mixed-domain retrieval	Δ	-1.82	-0.48	-0.67	2.84	1568

Table 18 covers the Basque CasiMédicos-Arg EU dev set (multiple-choice, 55 examples).

Table 18: Basque CasiMedicos EU dev ablation (55 examples, multiple-choice). Same metrics as Table 14.

#	Model	Experiment	SF	Quality ↑			Cost ↓	
				ROUGE-L	Cosim	BERT-F1	sec	tok
0a	Llama-3.1-8B-Instruct	Baseline LLM only	noSF	17.34	87.26	69.70	4.79	634
	Llama-3.1-8B-Instruct	Baseline LLM only	SF	14.47	86.65	68.63	10.13	1368
	Llama-3.1-8B-Instruct	Baseline LLM only	Δ	-2.87	-0.62	-1.07	5.34	734
0b	Latxa-Llama-3.1-8B-Instruct	Baseline LLM only	noSF	22.19	88.05	71.99	5.53	665
	Latxa-Llama-3.1-8B-Instruct	Baseline LLM only	SF	23.57	88.37	72.78	10.71	1392
	Latxa-Llama-3.1-8B-Instruct	Baseline LLM only	Δ	1.38	0.32	0.79	5.18	727
1a	Llama-3.1-8B-Instruct	e5 top 1	noSF	20.13	87.26	70.00	4.47	945
	Llama-3.1-8B-Instruct	e5 top 1	SF	19.81	87.91	71.36	9.22	1980
	Llama-3.1-8B-Instruct	e5 top 1	Δ	-0.33	0.66	1.36	4.74	1035
1b	Latxa-Llama-3.1-8B-Instruct	e5 top 1	noSF	16.63	84.87	66.59	2.61	868
	Latxa-Llama-3.1-8B-Instruct	e5 top 1	SF	20.19	86.14	68.09	6.08	1852
	Latxa-Llama-3.1-8B-Instruct	e5 top 1	Δ	3.56	1.27	1.49	3.47	983
2a	Llama-3.1-8B-Instruct	e5 top 3	noSF	26.88	88.20	73.31	4.26	1606
	Llama-3.1-8B-Instruct	e5 top 3	SF	27.35	88.92	74.17	8.73	3299
	Llama-3.1-8B-Instruct	e5 top 3	Δ	0.46	0.72	0.86	4.46	1693
2b	Latxa-Llama-3.1-8B-Instruct	e5 top 3	noSF	19.99	86.60	69.73	3.33	1568
	Latxa-Llama-3.1-8B-Instruct	e5 top 3	SF	20.99	86.59	69.01	7.10	3233
	Latxa-Llama-3.1-8B-Instruct	e5 top 3	Δ	1.01	-0.01	-0.72	3.76	1665
3a	Llama-3.1-8B-Instruct	e5 top 5	noSF	28.22	88.45	72.81	4.30	2248
	Llama-3.1-8B-Instruct	e5 top 5	SF	31.25	89.92	75.41	8.66	4577
	Llama-3.1-8B-Instruct	e5 top 5	Δ	3.03	1.47	2.60	4.36	2329
3b	Latxa-Llama-3.1-8B-Instruct	e5 top 5	noSF	23.30	87.33	71.00	3.66	2222
	Latxa-Llama-3.1-8B-Instruct	e5 top 5	SF	25.23	87.98	71.56	7.73	4539
	Latxa-Llama-3.1-8B-Instruct	e5 top 5	Δ	1.94	0.65	0.56	4.07	2317
4a	Llama-3.1-8B-Instruct	rerank top 1	noSF	17.60	86.80	69.34	4.84	916
	Llama-3.1-8B-Instruct	rerank top 1	SF	20.11	87.83	71.42	9.57	1922
	Llama-3.1-8B-Instruct	rerank top 1	Δ	2.51	1.03	2.08	4.73	1006
4b	Latxa-Llama-3.1-8B-Instruct	rerank top 1	noSF	12.74	84.42	66.40	3.17	849
	Latxa-Llama-3.1-8B-Instruct	rerank top 1	SF	16.06	85.32	67.65	7.31	1831
	Latxa-Llama-3.1-8B-Instruct	rerank top 1	Δ	3.31	0.90	1.26	4.14	982
5a	Llama-3.1-8B-Instruct	rerank top 3	noSF	21.60	87.69	70.66	4.90	1553
	Llama-3.1-8B-Instruct	rerank top 3	SF	25.74	88.88	73.42	9.21	3176
	Llama-3.1-8B-Instruct	rerank top 3	Δ	4.14	1.19	2.77	4.30	1623
5b	Latxa-Llama-3.1-8B-Instruct	rerank top 3	noSF	15.95	85.93	68.97	3.68	1502
	Latxa-Llama-3.1-8B-Instruct	rerank top 3	SF	19.04	86.47	68.34	7.42	3102
	Latxa-Llama-3.1-8B-Instruct	rerank top 3	Δ	3.10	0.54	-0.64	3.74	1600
6a	Llama-3.1-8B-Instruct	rerank top 5	noSF	27.02	88.74	73.25	4.77	2243
	Llama-3.1-8B-Instruct	rerank top 5	SF	25.94	89.18	73.67	9.29	4564
	Llama-3.1-8B-Instruct	rerank top 5	Δ	-1.08	0.44	0.42	4.51	2321
6b	Latxa-Llama-3.1-8B-Instruct	rerank top 5	noSF	20.21	86.96	70.67	4.11	2210
	Latxa-Llama-3.1-8B-Instruct	rerank top 5	SF	19.56	86.71	69.28	8.13	4512
	Latxa-Llama-3.1-8B-Instruct	rerank top 5	Δ	-0.65	-0.25	-1.39	4.02	2302
7a	Llama-3.1-8B-Instruct	3-shot, no RAG	noSF	24.12	89.11	74.02	4.21	1854
	Llama-3.1-8B-Instruct	3-shot, no RAG	SF	17.81	87.51	70.91	9.03	2566
	Llama-3.1-8B-Instruct	3-shot, no RAG	Δ	-6.31	-1.59	-3.10	4.82	712
7b	Latxa-Llama-3.1-8B-Instruct	3-shot, no RAG	noSF	25.18	88.78	74.34	3.75	1835
	Latxa-Llama-3.1-8B-Instruct	3-shot, no RAG	SF	25.48	89.06	74.48	7.58	2506
	Latxa-Llama-3.1-8B-Instruct	3-shot, no RAG	Δ	0.31	0.28	0.15	3.83	671
8a	Llama-3.1-8B-Instruct	3-shot + rerank top 5	noSF	29.32	89.39	75.04	4.66	3478
	Llama-3.1-8B-Instruct	3-shot + rerank top 5	SF	27.09	89.42	74.77	8.69	5780
	Llama-3.1-8B-Instruct	3-shot + rerank top 5	Δ	-2.22	0.04	-0.28	4.03	2302
⇒ 8b	Latxa-Llama-3.1-8B-Instruct	3-shot + rerank top 5	noSF	28.47	89.48	75.48	3.93	3451
	Latxa-Llama-3.1-8B-Instruct	3-shot + rerank top 5	SF	23.05	88.06	71.77	7.74	5745

Continued on next page

Table 18 (continued)

#	Model	Experiment	SF	Quality ↑			Cost ↓	
				ROUGE-L	Cosim	BERT-F1	sec	tok
	Latxa-Llama-3.1-8B-Instruct	3-shot + rerank top 5	Δ	-5.42	-1.43	-3.71	3.82	2294
9a	Llama-3.1-8B-Instruct	cross-domain retrieval	noSF	8.94	83.46	64.89	5.76	2037
	Llama-3.1-8B-Instruct	cross-domain retrieval	SF	11.49	84.06	67.17	9.37	4081
	Llama-3.1-8B-Instruct	cross-domain retrieval	Δ	2.55	0.60	2.29	3.62	2044
9b	Latxa-Llama-3.1-8B-Instruct	cross-domain retrieval	noSF	8.53	83.34	65.77	3.24	1936
	Latxa-Llama-3.1-8B-Instruct	cross-domain retrieval	SF	11.70	83.81	66.88	7.04	3987
	Latxa-Llama-3.1-8B-Instruct	cross-domain retrieval	Δ	3.17	0.47	1.11	3.79	2051
10a	Llama-3.1-8B-Instruct	mixed-domain retrieval	noSF	27.02	88.74	73.25	4.92	2243
	Llama-3.1-8B-Instruct	mixed-domain retrieval	SF	25.94	89.18	73.67	9.44	4564
	Llama-3.1-8B-Instruct	mixed-domain retrieval	Δ	-1.08	0.44	0.42	4.51	2321
10b	Latxa-Llama-3.1-8B-Instruct	mixed-domain retrieval	noSF	20.21	86.96	70.67	4.11	2210
	Latxa-Llama-3.1-8B-Instruct	mixed-domain retrieval	SF	19.56	86.71	69.28	8.13	4512
	Latxa-Llama-3.1-8B-Instruct	mixed-domain retrieval	Δ	-0.65	-0.25	-1.39	4.02	2302

Table 19 covers the Basque mixed dev set (161 examples).

Table 19: Basque SNS1064+CasiMedicos EU dev ablation (161 examples, mixed). Same metrics as Table 14.

#	Model	Experiment	SF	Quality ↑			Cost ↓	
				ROUGE-L	Cosim	BERT-F1	sec	tok
0a	Llama-3.1-8B-Instruct	Baseline LLM only	noSF	9.00	82.77	63.32	5.66	552
	Llama-3.1-8B-Instruct	Baseline LLM only	SF	8.03	82.66	64.43	11.35	1181
	Llama-3.1-8B-Instruct	Baseline LLM only	Δ	-0.97	-0.10	1.10	5.69	629
0b	Latxa-Llama-3.1-8B-Instruct	Baseline LLM only	noSF	11.30	83.24	65.84	5.87	560
	Latxa-Llama-3.1-8B-Instruct	Baseline LLM only	SF	11.61	83.29	66.08	11.46	1185
	Latxa-Llama-3.1-8B-Instruct	Baseline LLM only	Δ	0.31	0.05	0.24	5.59	625
1a	Llama-3.1-8B-Instruct	e5 top 1	noSF	18.00	85.79	70.28	3.78	711
	Llama-3.1-8B-Instruct	e5 top 1	SF	16.31	85.71	69.99	7.84	1511
	Llama-3.1-8B-Instruct	e5 top 1	Δ	-1.69	-0.09	-0.28	4.06	801
1b	Latxa-Llama-3.1-8B-Instruct	e5 top 1	noSF	19.66	85.82	70.48	2.49	657
	Latxa-Llama-3.1-8B-Instruct	e5 top 1	SF	18.68	85.64	69.94	6.01	1436
	Latxa-Llama-3.1-8B-Instruct	e5 top 1	Δ	-0.98	-0.18	-0.54	3.52	779
2a	Llama-3.1-8B-Instruct	e5 top 3	noSF	25.15	87.31	73.53	3.31	1166
	Llama-3.1-8B-Instruct	e5 top 3	SF	22.04	86.78	72.30	6.97	2426
	Llama-3.1-8B-Instruct	e5 top 3	Δ	-3.11	-0.53	-1.24	3.66	1259
2b	Latxa-Llama-3.1-8B-Instruct	e5 top 3	noSF	23.43	86.71	72.08	2.72	1142
	Latxa-Llama-3.1-8B-Instruct	e5 top 3	SF	24.29	86.79	72.09	5.92	2383
	Latxa-Llama-3.1-8B-Instruct	e5 top 3	Δ	0.86	0.08	0.00	3.20	1241
3a	Llama-3.1-8B-Instruct	e5 top 5	noSF	28.99	87.85	74.34	3.05	1653
	Llama-3.1-8B-Instruct	e5 top 5	SF	27.06	87.79	73.92	6.54	3403
	Llama-3.1-8B-Instruct	e5 top 5	Δ	-1.93	-0.06	-0.42	3.49	1750
3b	Latxa-Llama-3.1-8B-Instruct	e5 top 5	noSF	30.09	88.21	74.53	2.97	1650
	Latxa-Llama-3.1-8B-Instruct	e5 top 5	SF	28.90	88.04	74.25	5.97	3380
	Latxa-Llama-3.1-8B-Instruct	e5 top 5	Δ	-1.18	-0.17	-0.27	3.00	1730
4a	Llama-3.1-8B-Instruct	rerank top 1	noSF	18.95	86.12	70.71	4.37	745
	Llama-3.1-8B-Instruct	rerank top 1	SF	16.88	85.73	69.99	8.91	1587
	Llama-3.1-8B-Instruct	rerank top 1	Δ	-2.07	-0.39	-0.72	4.54	842
4b	Latxa-Llama-3.1-8B-Instruct	rerank top 1	noSF	19.20	85.94	70.45	3.43	705
	Latxa-Llama-3.1-8B-Instruct	rerank top 1	SF	20.01	86.10	70.90	7.34	1522
	Latxa-Llama-3.1-8B-Instruct	rerank top 1	Δ	0.81	0.16	0.46	3.92	817
5a	Llama-3.1-8B-Instruct	rerank top 3	noSF	26.20	87.70	73.32	4.24	1249
	Llama-3.1-8B-Instruct	rerank top 3	SF	24.63	87.42	73.04	8.29	2584
	Llama-3.1-8B-Instruct	rerank top 3	Δ	-1.57	-0.28	-0.28	4.05	1334

Continued on next page

Table 19 (continued)

#	Model	Experiment	SF	Quality ↑ ROUGE-L	Cosim	BERT-F1	Cost ↓ sec	tok
5b	Latxa-Llama-3.1-8B-Instruct	rerank top 3	noSF	25.98	87.37	73.33	3.54	1220
	Latxa-Llama-3.1-8B-Instruct	rerank top 3	SF	26.40	87.45	72.94	6.95	2528
	Latxa-Llama-3.1-8B-Instruct	rerank top 3	Δ	0.43	0.08	-0.39	3.42	1308
6a	Llama-3.1-8B-Instruct	rerank top 5	noSF	30.93	88.55	75.28	3.86	1753
	Llama-3.1-8B-Instruct	rerank top 5	SF	28.99	88.23	74.67	7.54	3589
	Llama-3.1-8B-Instruct	rerank top 5	Δ	-1.95	-0.32	-0.61	3.67	1836
6b	Latxa-Llama-3.1-8B-Instruct	rerank top 5	noSF	29.38	88.14	74.51	3.50	1739
	Latxa-Llama-3.1-8B-Instruct	rerank top 5	SF	27.96	87.73	73.59	6.74	3558
	Latxa-Llama-3.1-8B-Instruct	rerank top 5	Δ	-1.42	-0.40	-0.91	3.23	1819
7a	Llama-3.1-8B-Instruct	3-shot, no RAG	noSF	14.08	84.93	67.15	4.64	1392
	Llama-3.1-8B-Instruct	3-shot, no RAG	SF	10.30	83.37	65.32	10.06	2009
	Llama-3.1-8B-Instruct	3-shot, no RAG	Δ	-3.78	-1.56	-1.83	5.41	618
7b	Latxa-Llama-3.1-8B-Instruct	3-shot, no RAG	noSF	12.56	84.47	67.56	4.32	1378
	Latxa-Llama-3.1-8B-Instruct	3-shot, no RAG	SF	12.55	84.21	67.50	8.67	1952
	Latxa-Llama-3.1-8B-Instruct	3-shot, no RAG	Δ	-0.02	-0.26	-0.06	4.35	574
8a	Llama-3.1-8B-Instruct	3-shot + rerank top 5	noSF	27.93	87.69	74.19	3.74	2630
	Llama-3.1-8B-Instruct	3-shot + rerank top 5	SF	28.70	88.17	74.52	7.30	4462
	Llama-3.1-8B-Instruct	3-shot + rerank top 5	Δ	0.77	0.48	0.34	3.56	1832
⇒ 8b	Latxa-Llama-3.1-8B-Instruct	3-shot + rerank top 5	noSF	32.14	88.65	75.88	3.61	2626
	Latxa-Llama-3.1-8B-Instruct	3-shot + rerank top 5	SF	30.11	88.40	74.89	6.99	4450
	Latxa-Llama-3.1-8B-Instruct	3-shot + rerank top 5	Δ	-2.03	-0.25	-0.99	3.37	1824
9a	Llama-3.1-8B-Instruct	SNS-only retrieval	noSF	24.76	86.75	72.42	4.15	1683
	Llama-3.1-8B-Instruct	SNS-only retrieval	SF	24.05	86.48	72.45	7.52	3424
	Llama-3.1-8B-Instruct	SNS-only retrieval	Δ	-0.71	-0.26	0.03	3.36	1741
9b	Latxa-Llama-3.1-8B-Instruct	SNS-only retrieval	noSF	25.39	86.90	72.83	3.25	1645
	Latxa-Llama-3.1-8B-Instruct	SNS-only retrieval	SF	25.28	86.74	72.77	6.42	3379
	Latxa-Llama-3.1-8B-Instruct	SNS-only retrieval	Δ	-0.11	-0.16	-0.06	3.16	1733
10a	Llama-3.1-8B-Instruct	CasiMedicos-only retrieval	noSF	13.67	83.95	65.06	5.28	2263
	Llama-3.1-8B-Instruct	CasiMedicos-only retrieval	SF	13.32	84.17	66.88	10.12	4598
	Llama-3.1-8B-Instruct	CasiMedicos-only retrieval	Δ	-0.35	0.23	1.82	4.84	2335
10b	Latxa-Llama-3.1-8B-Instruct	CasiMedicos-only retrieval	noSF	11.62	83.54	65.50	4.22	2220
	Latxa-Llama-3.1-8B-Instruct	CasiMedicos-only retrieval	SF	10.08	83.01	63.02	8.95	4551
	Latxa-Llama-3.1-8B-Instruct	CasiMedicos-only retrieval	Δ	-1.54	-0.53	-2.48	4.73	2331

7 Discussion

7.1 Effect of retrieval and reranking

7.2 Effect of few-shot prompting

7.3 Effect of self-feedback

7.4 Effect of thinking mode (Qwen3.5-9B)

7.5 Qwen3.5-4B vs. Qwen3.5-9B: quality-cost tradeoff

To assess whether the larger model is worth the extra compute, Table 20 compares the best configuration found for each model size on each Spanish dataset. The best configuration per model was identified by selecting the highest ROUGE-L result across the full ablation grid.

Dataset	Model	Best config	ROUGE-L	Cosim	BERT-F1	sec	tok
SNS-1064	4B	3-shot + rerank5, noSF	38.69	89.35	78.81	3.49	1983
	9B	3-shot + rerank5, noSF	38.68	89.61	78.91	4.10	1989
		Δ (9B − 4B)	−0.01	+0.26	+0.10	+0.61	+6
CasiMédicos-Arg	4B	3-shot + rerank5, SF	45.30	92.23	81.50	7.51	4913
	9B	3-shot, no RAG, SF	51.55	93.71	83.28	14.36	2286
		Δ (9B − 4B)	+6.25	+1.48	+1.78	+6.85	−2627
Mixed	4B	3-shot + rerank5, SF	39.46	89.60	78.86	5.45	3896
	9B	rerank5, SF	41.08	90.12	79.44	7.07	3016
		Δ (9B − 4B)	+1.62	+0.52	+0.58	+1.62	−880

Table 20: Best-configuration quality and cost comparison of Qwen3.5-4B vs. Qwen3.5-9B (Spanish dev). Δ = 9B best − 4B best.

The best configuration is identical for both model sizes on SNS-1064 (**3-shot + rerank5, no_think, noSF**), confirming that open-answer clinical QA converges on the same retrieval strategy regardless of model size. On CasiMédicos-Arg the best configs differ: 4B peaks at 3-shot + rerank5 with SF (45.30 ROUGE-L), while 9B reaches its maximum with 3-shot, no RAG, with SF (51.55 ROUGE-L) — a +6.25 point gap. On the Mixed dataset both models again prefer SF, though 4B benefits more from reranking (3-shot + rerank5) while 9B is sufficient with rerank5 alone; the quality gap narrows to +1.62 ROUGE-L.

Regarding cost, SNS-1064 shows a straightforward tradeoff: 9B is 17% slower (+0.61 s/sample) at virtually equal token consumption, for a quality gain below 0.1 ROUGE-L points. On CasiMédicos-Arg the cost picture is inverted: 9B's best config (no RAG) is much cheaper in tokens (−2,627 tok/sample) but nearly twice as slow in wall time, because it generates longer outputs without the grounding that RAG provides to 4B. Overall, going from 4B to 9B yields an average quality gain of +6.0% ROUGE-L, +0.8% cosine similarity, and +1.0% BERT-F1 across best configs, at an average latency increase of +46% — driven almost entirely by the CasiMédicos-Arg gap where the two models take very different optimal paths. For open-answer clinical QA (SNS-1064), 4B matches 9B at lower cost and should be preferred; for multiple-choice tasks (CasiMédicos-Arg), 9B's substantially higher ceiling justifies the extra latency.

7.6 Spanish vs. Basque

8 Conclusions

References

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *The Twelfth International Conference on Learning Representations*, Vienna, Austria, 2024. OpenReview.net. URL https://openreview.net/forum?id=hSyW5go0v8.

Akari Asai, Jacqueline He, Rulin Shao, Weijia Shi, Amanpreet Singh, Joseph Chee Chang, Kyle Lo, Luca Soldaini, Sergey Feldman, Mike D'Arcy, David Wadden, Matt Latzke, Jenna Sparks, Jena D. Hwang, Varsha Kishore, Minyang Tian, Pan Ji, Shengyan Liu, Hao Tong, Bohao Wu, Yanyu Xiong, Luke Zettlemoyer, Graham Neubig, Daniel S. Weld, Doug Downey, Wen-Tau Yih, Pang Wei Koh, and Hannaneh Hajishirzi. Synthesizing scientific literature with retrieval-augmented language models. *Nature*, 650(8103):857–863, February 2026. doi: 10.1038/s41586-025-10072-4. URL https://www.nature.com/articles/s41586-025-10072-4.

Geek Software GmbH. PDF24 Tools, 2026. URL https://tools.pdf24.org/en/pdf-to-txt#s=1773652138058. [Online tool for PDF conversion and processing].

GuíaSalud. Portal GuíaSalud: Catálogo de Guías de Práctica Clínica en el Sistema Nacional de Salud (SNS), 2026. URL https://portal.guiasalud.es/gpc/. Accessed: 2026-04-23.

Hyunjae Kim, Jiwoong Sohn, Aidan Gilson, Nicholas Cochran-Caggiano, Serina Applebaum, Heeju Jin, Seihee Park, Yujin Park, Jiyeong Park, Seoyoung Choi, Brittany Alexandra Herrera Contreras, Thomas Huang, Jaehoon Yun, Ethan F. Wei, Roy Jiang, Leah Colucci, Eric Lai, Amisha Dave, Tuo Guo, Maxwell B. Singer, Yonghoe Koo, Ron A. Adelman, James Zou, Andrew Taylor, Arman Cohan, Hua Xu, and Qingyu Chen. Rethinking retrieval-augmented generation for medicine: A large-scale, systematic expert evaluation and practical insights. *arXiv preprint arXiv:2511.06738*, 2025. URL https://arxiv.org/abs/2511.06738.

Sichu Liang, Linhai Zhang, Hongyu Zhu, Wenwen Wang, Yulan He, and Deyu Zhou. RGAR: Recurrence generation-augmented retrieval for factual-aware medical question answering. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 4006–4033, Suzhou, China, November 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-emnlp.214. URL https://aclanthology.org/2025.findings-emnlp.214/.

Ministerio de Sanidad. Guía de Práctica Clínica para el Tratamiento del Trastorno de Ansiedad Generalizada en Atención Primaria, 2024a. URL https://doi.org/10.46995/gpc_641. Grupo de trabajo de la Guía de Práctica Clínica para el Tratamiento del Trastorno de Ansiedad Generalizada en Atención Primaria. Servicio de Evaluación del Servicio Canario de la Salud. Guías de Práctica Clínica en el SNS.

Ministerio de Sanidad. Guía de Práctica Clínica sobre Diabetes Mellitus Tipo 1, 2026. URL https://doi.org/10.46995/gpc_659. Grupo de trabajo de la Guía de Práctica Clínica sobre Diabetes Mellitus Tipo 1. Servicio de Evaluación del Servicio Canario de Salud. Guías de Práctica Clínica en el SNS.

Ministerio de Sanidad. Guía de Práctica Clínica sobre el Manejo del Ictus en Atención Primaria, 2024b. URL https://doi.org/10.46995/gpc_635. Grupo de trabajo de la Guía de Práctica Clínica sobre el Manejo del Ictus en Atención Primaria. Osteba/BIOEF. Guías de Práctica Clínica en el SNS.

Ministerio de Sanidad. Guía de Práctica Clínica sobre atención paliativa al adulto en situación de últimos días, 2021. URL https://doi.org/10.46995/gpc_612. Grupo de trabajo de la Guía de Práctica Clínica sobre atención paliativa al adulto en situación de últimos días. Agencia de Conocimiento en Salud (ACIS). Unidad de Asesoramiento Científico-técnico, Avalia-t. Guías de Práctica Clínica en el SNS.

Ministerio de Sanidad. Guía de Práctica Clínica sobre Cuidados Paliativos en Pediatría, 2022. URL https://doi.org/10.46995/gpc_618. Grupo de trabajo de la Guía de Práctica Clínica sobre Cuidados Paliativos en Pediatría. Instituto Aragonés de Ciencias de la Salud. Guías de Práctica Clínica en el SNS.

Ministerio de Sanidad. Guía de Práctica Clínica sobre prevención secundaria de ictus. Actualización, 2023. URL https://doi.org/10.46995/gpc_625. Grupo de trabajo de la Guía de Práctica Clínica sobre prevención secundaria de ictus. Actualización. AETSA-Evaluación de Tecnologías Sanitarias de Andalucía. Guías de Práctica Clínica en el SNS.

Shintaro Ozaki, Yuta Kato, Siyuan Feng, Masayo Tomita, Kazuki Hayashi, Wataru Hashimoto, Ryoma Obara, Masafumi Oyamada, Katsuhiko Hayashi, Hidetaka Kamigaito, and Taro Watanabe. Understanding the impact of confidence in retrieval augmented generation: A case study in the medical domain. In *Proceedings of the 24th Workshop on Biomedical Language Processing*, pages 1–17, Vienna, Austria, August 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.bionlp-1.1. URL https://aclanthology.org/2025.bionlp-1.1/.

Aswini Sivakumar, Vijayan Sugumaran, and Yao Qiang. RAG-X: Systematic diagnosis of retrieval-augmented generation for medical question answering. *arXiv preprint arXiv:2603.03541*, 2026. URL https://arxiv.org/abs/2603.03541.

Ekaterina Sviridova, Anar Yeginbergen, Ainara Estarrona, Elena Cabrio, Serena Villata, and Rodrigo Agerri. CasiMedicos-arg: A medical question answering dataset annotated with explanatory argumentative structures. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 18463–18475, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.1026.

Guangzhi Xiong, Qiao Jin, Zhiyong Lu, and Aidong Zhang. Benchmarking retrieval-augmented generation for medicine. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 6233–6251, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.372. URL https://aclanthology.org/2024.findings-acl.372/.